D2.3 Water potential

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d2-3.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 690–692 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Water potential

D2.3.8 What water potential is

p. 690

Water potential measures the stored energy held by a given volume of water, in kPa relative to pure water, which is set at 0 kPa. Solutes form hydrogen bonds with water molecules and lower their freedom to move, so solutions have negative water potentials.

D2.3.9 Direction of movement

p. 691

Water moves from higher (less negative) to lower (more negative) water potential, until the two are equal.

D2.3.10 Solute and pressure potential

pp. 691–692

Using the equation

D2.3.10

Cell	Solute potential / kPa	Pressure potential / kPa	Water potential / kPa
Turgid cell in pure water	−700	+700	0
Leaf cell, sunny day	-1000	+400	-600
Flaccid cell	-900	0	-900
Plasmolysed cell	-1200	0	-1200

Try it
Solute potential -800 kPa, pressure potential +300 kPa. Calculate the water potential.

Then
It is put in a solution at -400 kPa. Does it gain or lose water?

D2.3 Water potential (HL) – Textbook pp. 691–692 13

Water potential = solute potential + pressure potential. Solute potential is zero or negative. Pressure potential is usually positive in cells with walls, as the protoplast pushes on the wall, but negative in xylem under tension.

D2.3.11 Water movement in plant tissue

p. 692

In a hypotonic solution, water enters: solute potential becomes less negative and pressure potential rises until the cell is turgid. In a hypertonic solution, water leaves: solute potential becomes more negative and pressure potential falls to zero, and the cell may plasmolyse.

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